PS 2010: 5. Random Variable 1

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- You all did very well for the assignment 1 and 2
- Middle term exam next week
- All you need to know is in slides and homework assignment
- Extra Online Material
  - Introduction to Probability
  - Foundations of Statistics with R

Agenda

- Expectation and Variance
- Distribution of discrete random variable
- Distribution of continuous random variable

Random Variable

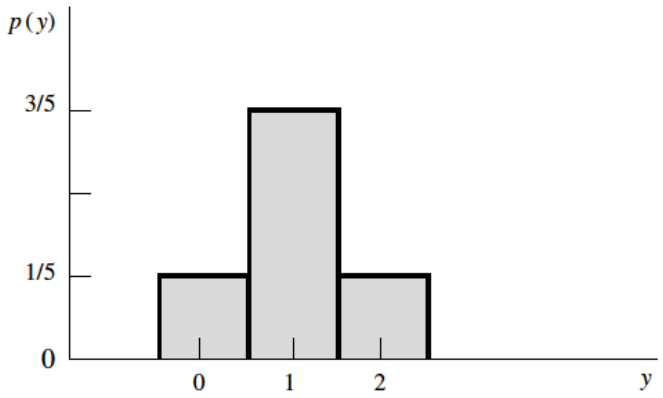
- **Random Variable** assigns a numeric value to each event of the experiment.
- Values need to be mutually exclusive and exhaustive events
- Two types of random variables:
  - Discrete random variable: takes a finite or at most countably infinite number of distinct values
  - Continuous: uncountably infinite number of values.

Probability Distribution for Discrete Random Variable

- Probability that Y takes on the value y is $P(Y = y)$ or $p(y)$
- Probability distribution for a discrete variable provides $p(y) = P(Y = y)$ for all y
- For discrete random variable, it is also **probability mass function, pmf**

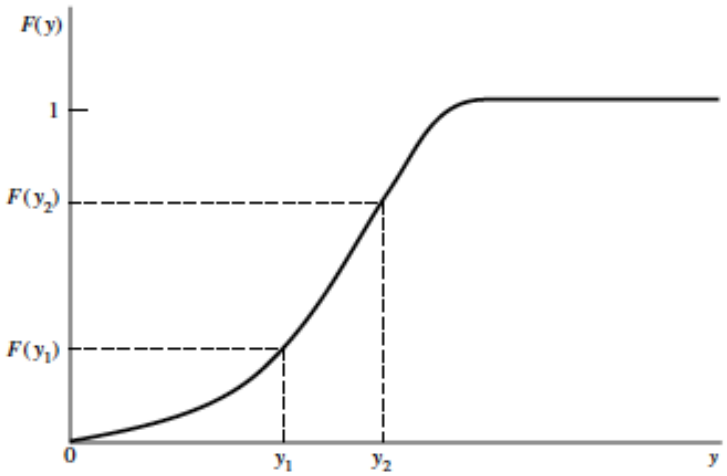
Table 3.1 Probability distribution for Example 3.1

y	$p(y)$
0	1/5
1	3/5
2	1/5



Probability Distribution for Continuous Random Variable

- Probability that Y takes on a specific value y is $P(Y = y) = p(y) = 0$
- Probability distribution for a continuous variable denoted by $F(y)$ is such that $F(y) = P(Y \leq y)$ for all $-\infty < y < \infty$
- The function $p(y)$ is **probability density function, pdf** for continuous variable



Cumulative Distribution Function

- The **cumulative distribution function** $F(y)$ of a discrete random variable y with probability distribution $f(x)$ is
$$F(y) = P(Y \leq y) = \sum_{t \leq y} f(y)$$
- The cumulative distribution function $F(y)$ of a continuous random variable y with probability distribution $f(x)$ is
$$F(y) = P(Y \leq y) = \int_{-\infty}^{y} f(t)dt$$
$$\text{for all } -\infty < y < \infty$$
- $\int_{-\infty}^{\infty} f(x)dx = 1$

$f(x)$ is the probability mass function or probability density function. $f(x) = p(x)$ just different notation.

$$P(a < X < b) = \int_a^b f(x)\, dx.$$

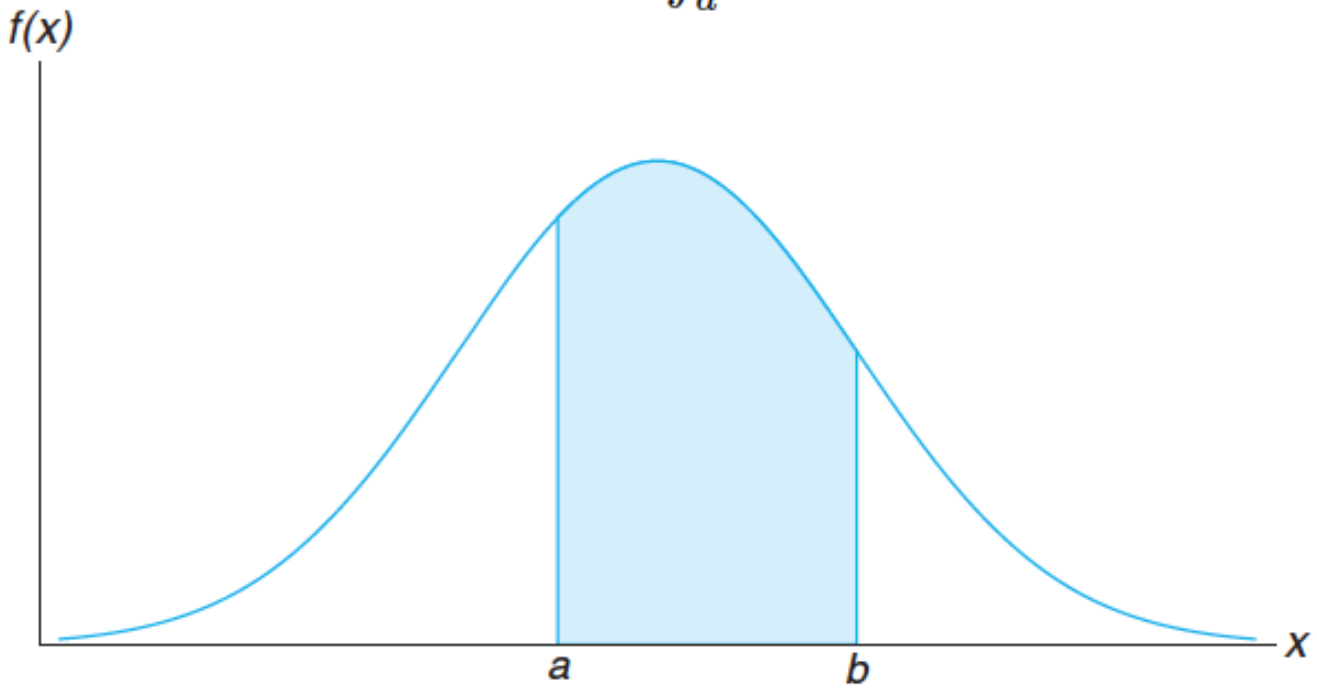


Figure 3.5: $P(a < X < b)$.

Example

In a political survey, it is found that 50% of respondents strongly support a particular policy aimed at enhancing national security

- Find a formula for the probability distribution of the number of respondents who strongly support the policy among the next 4 individuals surveyed.
- Find the cumulative distribution function of the random variable

Example

For a continuous random variable representing political campaign contributions, the probability density function is given by:

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

- Find the probability that a contributor will make campaign contributions over the entire possible range, $P(-\infty < X \leq \infty)$
- Find the probability that a contributor will make campaign contributions between \$0 and \$1, $P(0 < X \leq 1)$

Expectation

- Y is a discrete random variable, then the expected value of Y denoted as $E(Y)$ is $\mu = E(y) = \sum yf(y)$
- If Y is a continuous variable, then $\mu = E(y) = \int_{-\infty}^{\infty} yf(y)$
- Expectation represent the mean of random variables
- Let Y be a random variable with pdf f(y), the expected value of the discrete random variable g(Y) is

$$\mu_{g(Y)} = E[g(Y)] = \sum_y g(y)f(y)$$

- The expected value of the continuous random variable is

$$\mu_{g(Y)} = E[g(Y)] = \int_{-\infty}^{\infty} g(y)f(y)dy$$

Example

In a political poll, a surveyor selects a random sample of 3 individuals from a group of 7 potential voters. Among the group, 4 individuals support a particular candidate, while 3 individuals do not. Calculate the expected number of individuals in this sample who support the candidate.

Example

Let X be the random variable that represents the tenure in months of a particular government official in office. The probability density function is given by:

$$f(x) = \begin{cases} \frac{20{,}0}{x^3}, & x > 100 \\ 0, & \text{elsewhere.} \end{cases}$$

Calculate the expected tenure of this type of government official in months.

Example

Suppose that the number of voters X who arrive at a polling station between 4 : 00 P.M. and 5:00 P.M. on any sunny Friday has the following probability distribution:

x	4	5	6	7	8	9
$P(X = x)$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{6}$

Let $g(X) = 2X - 1$ represent the number of voters who express their support for a particular political candidate. Find the expected number of voters who support the candidate during this particular time period.

Example

Let X represent the random variable that measures the level of public trust in a political institution. The density function is given by:

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Calculate the expected change in public trust, denoted by $g(X) = 4X + 3$, due to a certain political event or policy change.

Rules of Expectation

Let X and Y be random variables, and a and b be arbitrary constants. Then

- $E(a) = a$
- $E(aX) = aE(X)$
- $E(aX + b) = aE(X) + b$
- $E(aX + bY) = aE(X) + bE(Y)$
- If X and Y are independent, then $E(XY) = E(X)E(Y)$, but generally, $E(XY) \neq E(X)E(Y)$

Example

A busy student must complete 3 problem sets before doing laundry. Each problem set requires 1 day with probability 2/3 and 2 days with probability 1/3. Let B be the number of days a busy student delays laundry. What is $E(B)$?

Variance

Definition: If X is a random variable with mean $E(X) = \mu$, the variance of a random variable X is defined to be the expected value of $(X - \mu)^2$:

$$V(X) = E[(X - \mu)^2]$$

The square root of X is called the standard deviation

Example: The probability distribution for a random variable Y is given in the table below. Find the mean, variance, and standard deviation of Y.

y	$p(y)$
0	1/8
1	1/4
2	3/8
3	1/4

Variance

$$
\begin{aligned}
V(X) &= E[(X - E(X))^2] \\
&= E[X^2 - 2X * E(X) + \{E(X)\}^2] \\
&= E(X^2) - 2E(X)E(X) + \{E(X)\}^2 \\
&= E(X^2) - 2\{E(X)\}^2 + \{E(X)\}^2 \\
&= E(X^2) - \{E(X)\}^2
\end{aligned}
$$

where we call $E(X)$ as the first moment, and $E(X^2)$ as the second moment

Example: A pdf $f(x) = \frac{1}{b-a}$ at interval [a, b], calculate the variance of random variable X

Rules of Variance

Let X and Y be random variables, and a and b be arbitrary constants. Then

- $V(a) = 0$
- $V(aX) = a^2 V(X)$
- $V(X + b) = V(X)$
- $V(aX + b) = a^2 V(X)$
- If X and Y are independent, $V(X + Y) = V(X) + V(Y)$

Distribution

- Distribution for discrete random variable
  - Bernoulli and Binomial
  - Negative Binomial
  - Poisson
- Distribution for continuous random variable
  - Uniform
  - Normal
  - Chi-square
  - Student t

Bernoulli and Binomial

- A random variable that takes two distinct values is called a **Bernoulli Random Variable**
- Success equals to 1 with probability p, and fail 0 with 1-p
- Example: flip a coin, election, classification
- **Binomial Random Variable** is the sum of n independently and identically distributed Bernoulli random variables.
- The probability distribution of this random variable is called binomial distribution.
- Example: conflict early warning system for the next ten times

Binomial

The probability mass function (PMF) of a binomial random variable with success probability p and n trials is given by

$$f(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}.$$

The cumulative distribution function (CDF) can be written as

$$F(x) = P(X \leq x) = \sum_{k=0}^{x} \binom{n}{k} p^k (1-p)^{n-k},$$

for $x = 0, 1, \ldots, n$.

Example

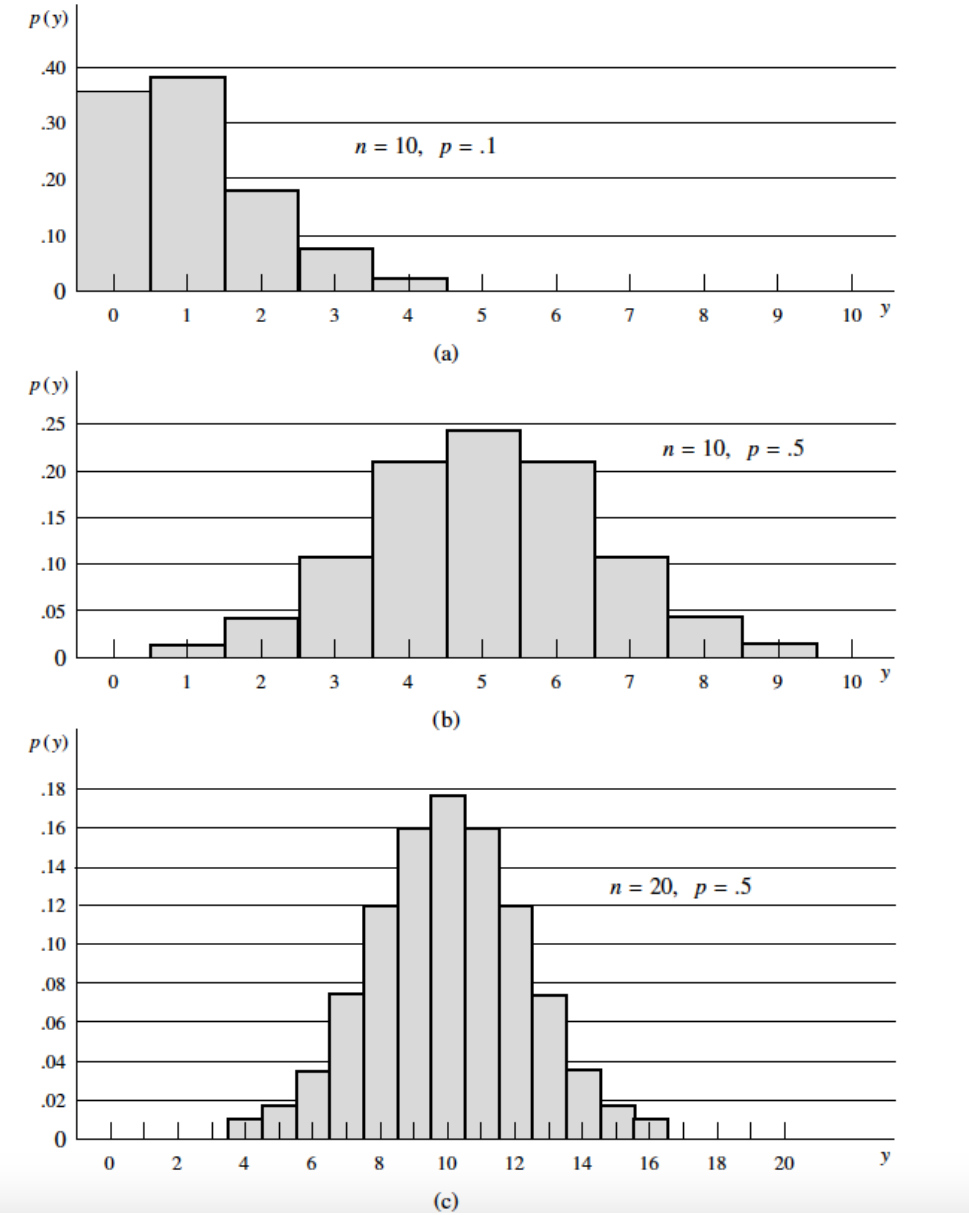
- In a hypothetical political election scenario, the probability of a candidate winning a specific constituency is 3/4. What is the probability that exactly 2 out of the next 4 constituencies in an election will be won by this candidate?
- 49% of Americans likes Trump, if we randomly select 10 people from public, what is the probability of at least 9 people like Trump
- In a political campaign, candidates often make various claims about their policy proposals. However, candidates make inaccurate claims 5% of the time. Suppose a fact-checker randomly examines three of these claims. Determine the probability distribution for Y, which represents the number of inaccurate claims detected by the fact-checker.

Mean and Variance of Binomial

Let X be a binomial random variable based on n trials and success probability p, then

$$\mu = E(X) = np \text{ and } \sigma^2 = V(X) = np(1-p)$$

Binomial Distribution



Negative Binomial

- Negative binomial also count events, but this time we care about the fixed number of success occur
- The number of X trials required to produce k successes is a negative binomial random variable
- A binomial random variable X has the following pdf:

$$f(x) = P(X = x) = \begin{pmatrix} x-1 \\ k-1 \end{pmatrix} p^k (1-p)^{x-k}.$$

- And the mean $\mu = E(X) = \frac{K}{p}$, and $\sigma^2 = V(X) = \frac{k(1-p)}{p^2}$

Example

- A conflict early warning system can predict successfully 55% percent of potential conflicts in the experimental stage. Now it handed over to another team for testing, and it can pass the testing if the system can make 4 success predictions in 7 testings.
- What is the probability that system pass the test at the 6th test
- What is the probability that the system pass the test

Example

In a political survey, a pollster is faced with a series of questions about political preferences, with three possible responses: "Conservative," "Liberal," and "Moderate." The pollster knows that 60% of respondents identify as Conservative, 30% as Liberal, and 10% as Moderate.

- Determine the probability that the pollster encounters the fifth Conservative respondent when surveying eight individuals.
- Calculate the probability that the pollster encounters 5 Conservative respondents in a sample of 8 individuals.

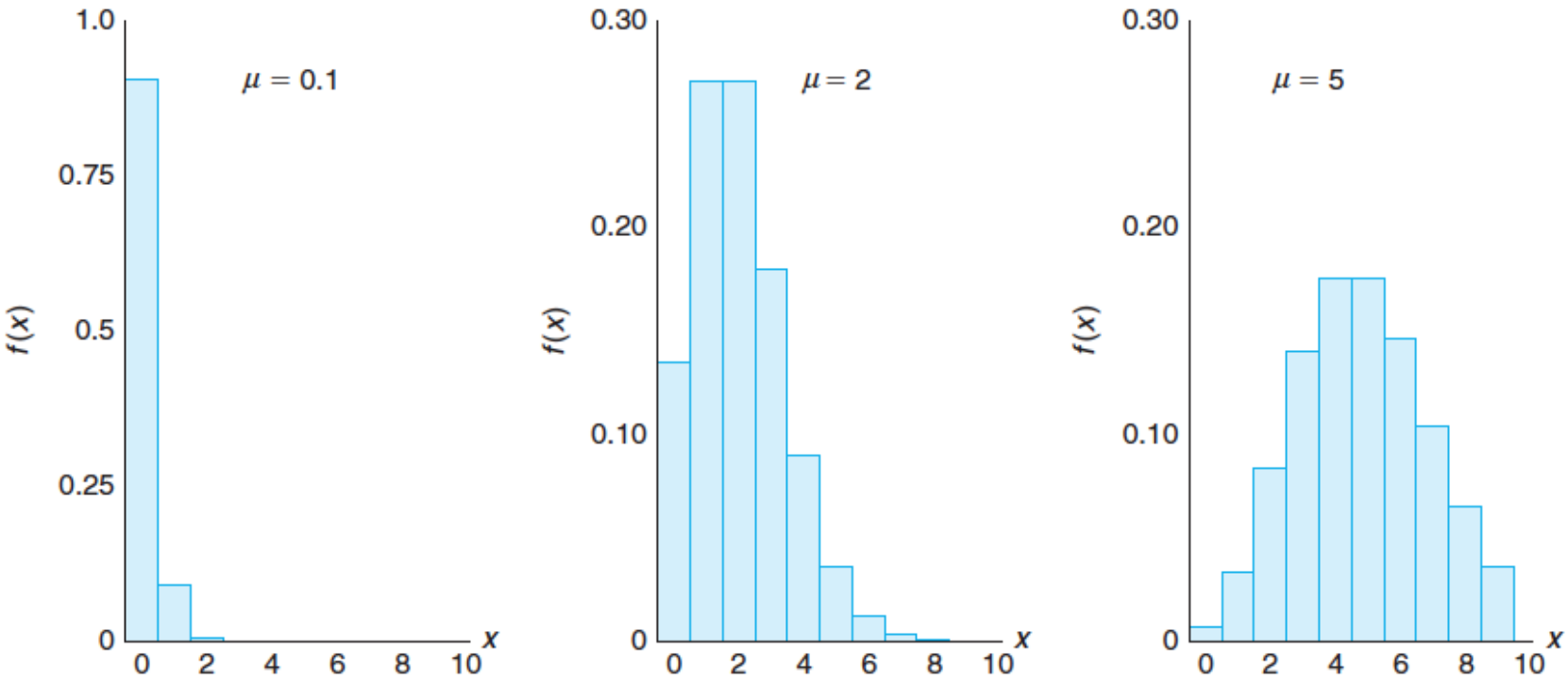
Poisson Distribution

- The number X of events occurring in one time interval is called a Poisson random variable
- Example: conflicts happening in one year, # of people dead due to Covid in a day
- The Poisson distribution for random variable X with mean λ, has the following pdf:

$$f(x) = p(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

- The means of Poisson distribution is $\mu = \lambda$ and $V(x) = \lambda$

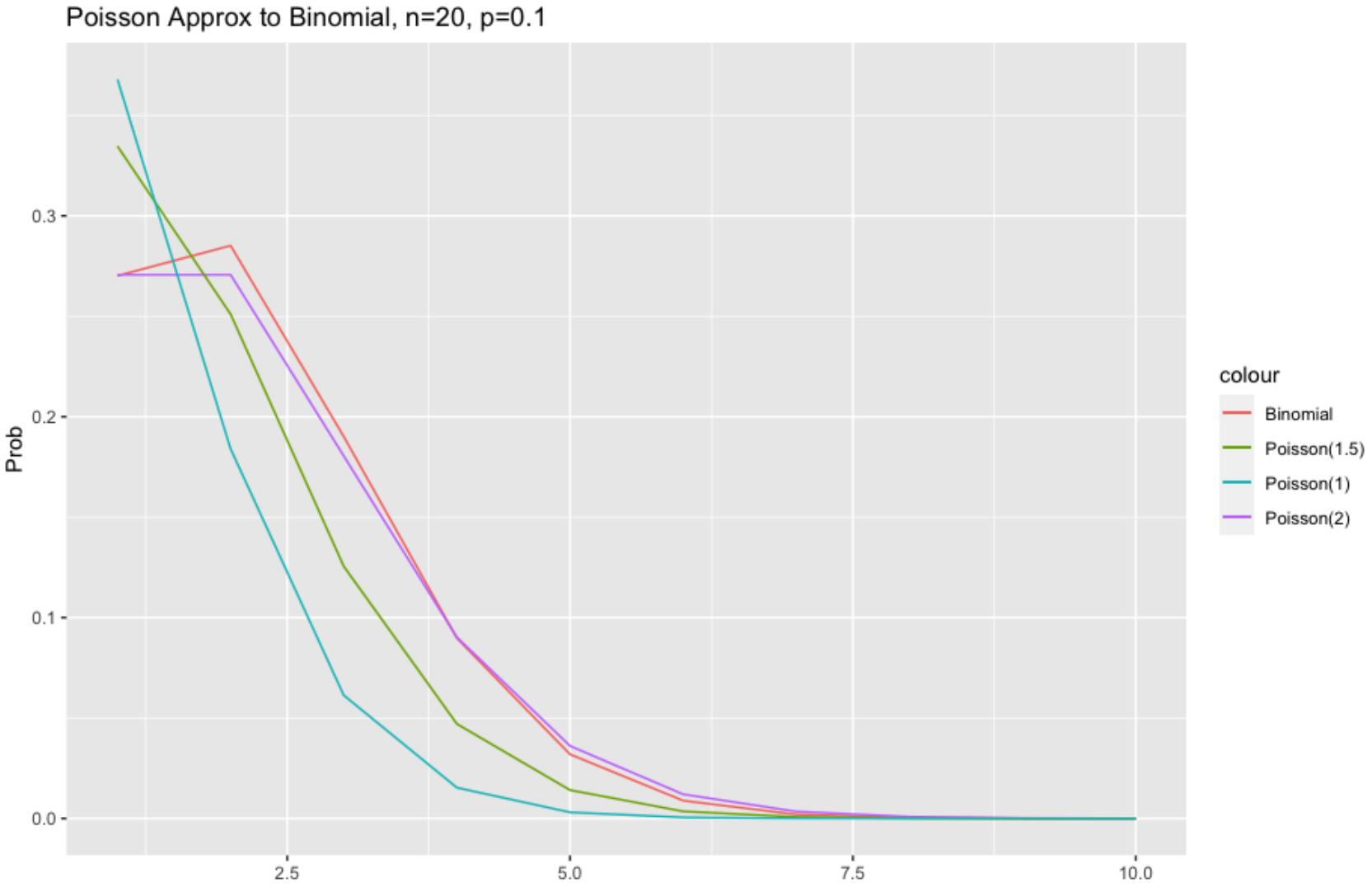
Poisson Example



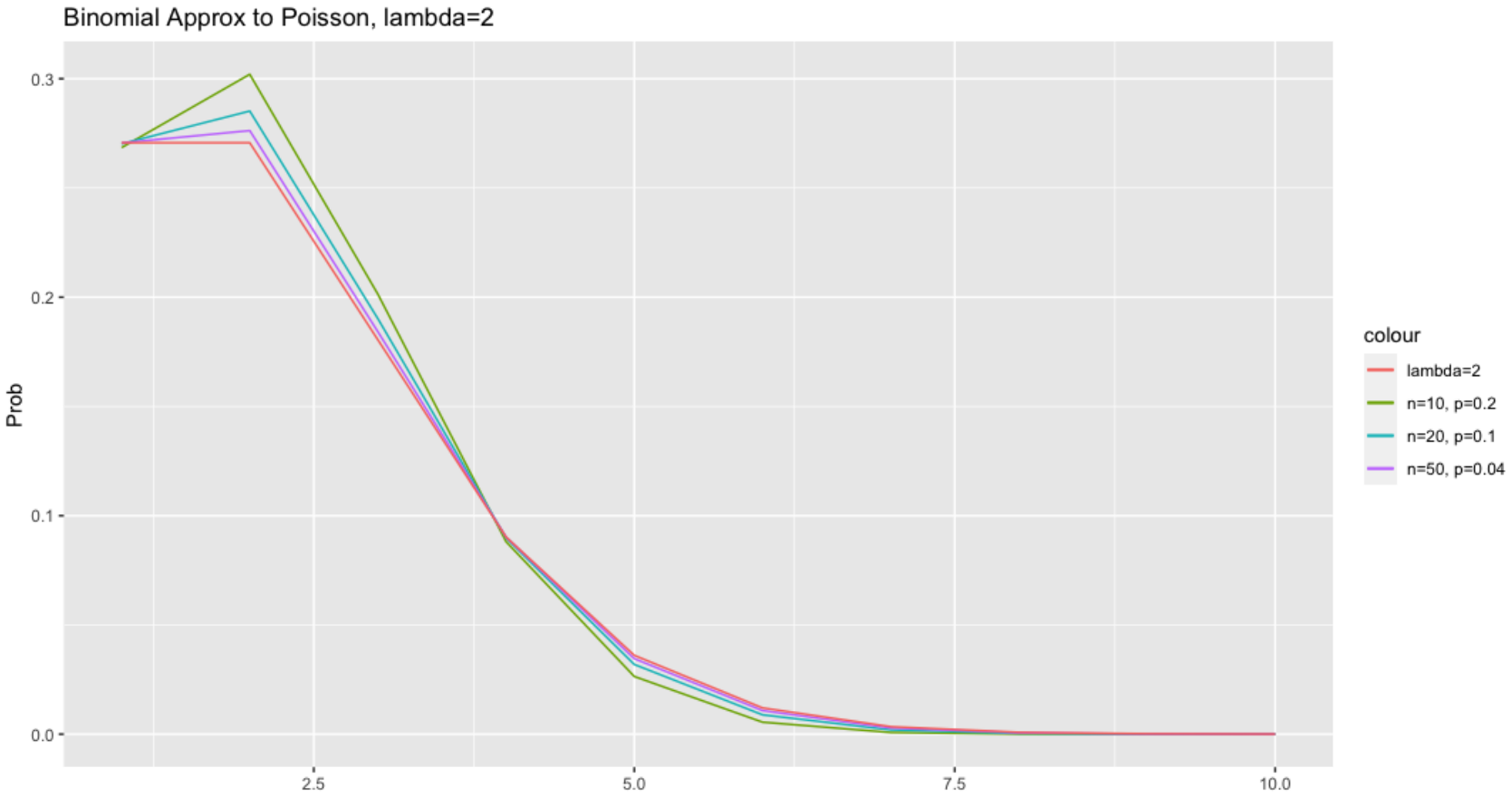
Example

- The average conflict of a year is 5, what is the probability that 6 conflicts happened in a year?
- The average number of bills passed by congress is 36 a year, what is the probability that 45 bills passed in a year?

Approximation of Binomial Distribution by a Poisson Distribution



Approximation of Poisson Distribution by a Binomial Distribution



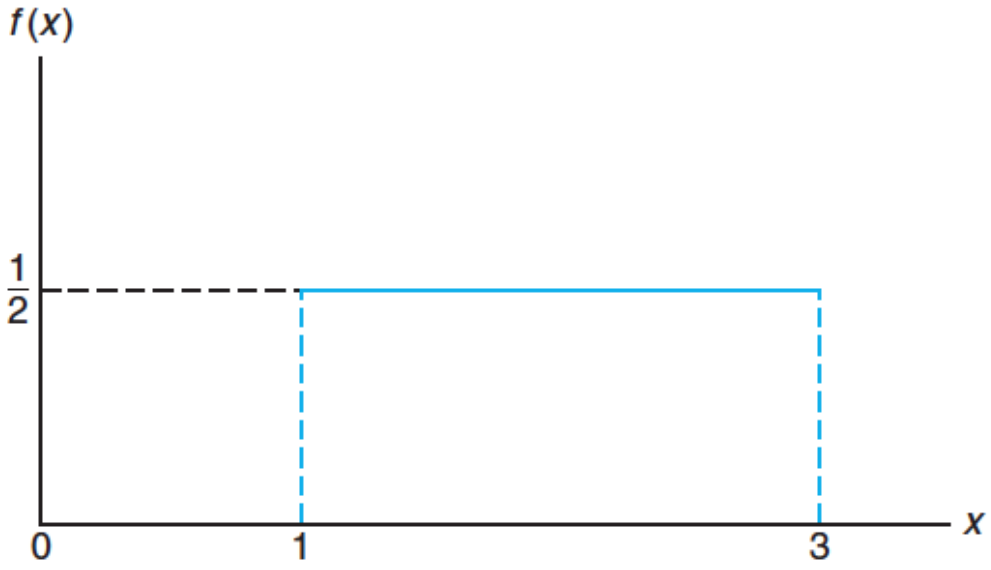
Uniform Distribution

- A very simple continuous probability distribution
- Pdf for a uniform random variable X on the interval [a,b]:

$$f(x) = \begin{cases} \frac{1}{B-A}, & A \leq x \leq B \\ 0, & \text{elsewhere} \end{cases}$$

- The density function forms a rectangle with base B-A and constant height $\frac{1}{B-A}$
- The mean of uniform random variable is $\mu = \frac{A+B}{2}$ and $\sigma^2 = \frac{(B-A)^2}{12}$

Uniform Distribution



Example

Hillman library require students to reserve study room for no more than 4 hours per person. In fact, it can be assumed that the length X of a reservation has a uniform distribution on the interval [0, 4].

- What is the probability density function?
- What is the probability that any given conference lasts at least 3 hours?

Normal Distribution

- Normal or Gaussian distribution takes the value from $-\infty$ to ∞
- The density function of a normal random variable X, with mean μ and variance σ^2 is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \ -\infty \leq x \leq \infty$$

- There is no analytical tractable form of cdf for a normal distribution.
- However, you can always calculate the probability between interval [a,b] as the area under the normal distribution curve

Standard Normal

- X is normal random variable with mean μ and standard deviation σ, denoted as $N(\mu, \sigma^2)$
- If a normal distribution has mean 0 and variance 1, then it is called **standard normal distribution**
- If c is a constant number, then $\mathrm{Z} = \mathrm{X} + \mathrm{c}$ is also a normal distribution with $\mathrm{Z} \sim N(\mu + c, \sigma^2)$
- If $\mathrm{Z} = \mathrm{cX}$, then $\mathrm{Z} \sim N(c\mu, (c\sigma)^2)$
- $z - score = \frac{X-\mu}{\sigma} \sim N(0, 1)$

Standard Normal

- For a standard normal, 1 standard deviation from the mean covers 68% of the observations

$$P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68$$

- 2 standard deviations from the mean covers 95%

$$P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$$

- z-score locates a point by how many in units of the standard deviation from the mean.
  Very useful when construct confidence interval and hypothesis testing
- A simple way to do standardization for variables

Example

- Given that X has a normal distribution with $\mu = 300$ and $\sigma = 50$, find the probability that X assumes a value greater than 362.
- Suppose the final grade for this class is normally distributed with $\mu = 75$ and $\sigma = 10$, what fraction of the scores lies between 80 and 90?
- If your instructor want 90% of students get an A and the rest of them are Bs, how should the instructor set the cutoff point?

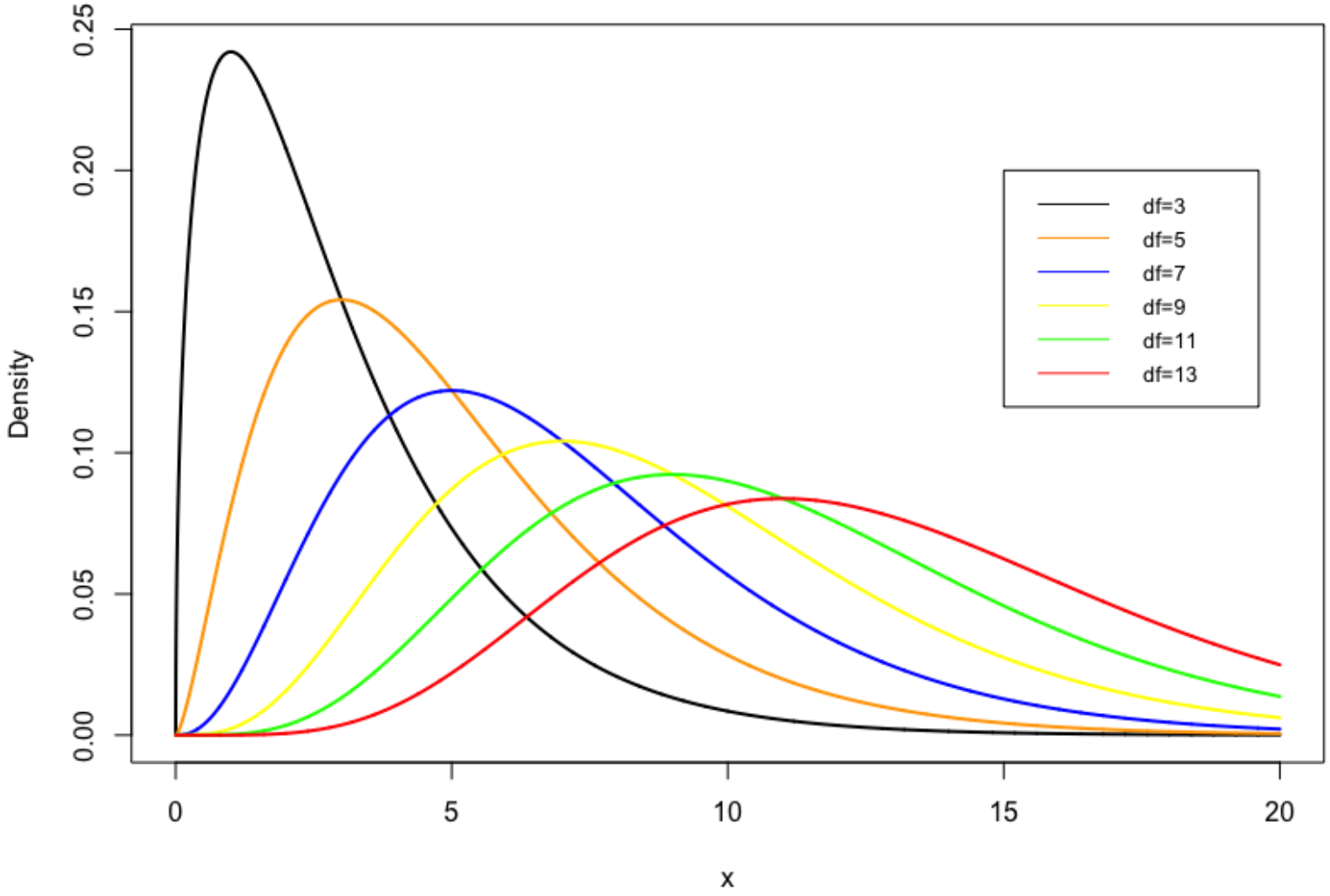
Chi-square

Several other commonly seen distributions:

- Chi-square, χ^2, distribution: If $Z_1, \ldots, Z_k$ are independent standard normal random variables, then

$$X^2 = Z_1^2 + \cdots + Z_k^2$$

  has a χ^2 distribution with k degrees of freedom.
- χ^2 distribution converges to normal distribution
- Useful for hypothesis testing for multivariate

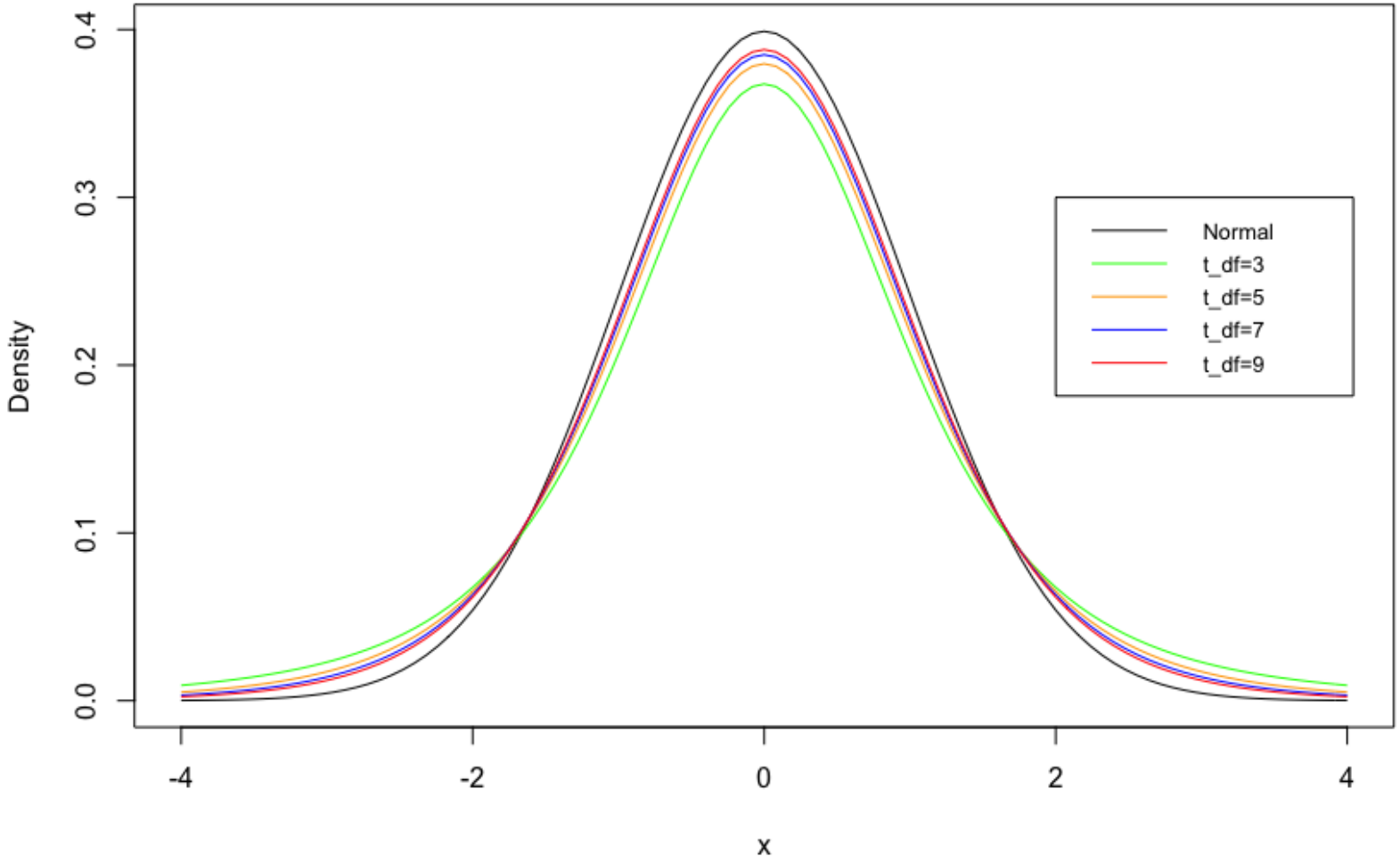

Density
0.00
0.05
0.10
0.15
0.20
0.25
0
5
10
15
20
x
df=3
df=5
df=7
df=9
df=11
df=13

Student t

If Z is a standard normal random variable and if X^2 is a χ^2 random variable with k degrees of freedom, then

$$T = \frac{Z}{\sqrt{X^2/k}}$$

has a t distribution with k degrees of freedom.

- t densities are symmetric, bell-shaped, and centered at 0 just like the standard normal density, but are more spread out (higher variance).
- As the degrees of freedom increases, the t distributions converge to the standard normal.
- t distributions will be useful for statistical inference for one or more populations of quantitative variables.

t distribution